

13. The equation of the tangent to the parabola with parametric equations $x = 2t^2$, $y = 4t$ at the point with parameter $t = 3$ is :

(A) $2y = x + 6$.

(B) $y = x + 6$.

(C) $3y = x + 18$.

(D) $y = x + 18$.

14. Identify the curve from the equation $7x^2 - 9y^2 = 36$.

(A) A parabola.

(B) An ellipse.

(C) A cycle.

(D) A hyperbola.

15. The matrix of the non-degenerate conic $x^2 + 8xy + 16y^2 - x + 8y - 12 = 0$:

(A) $\begin{bmatrix} 1 & 4 \\ 4 & 16 \end{bmatrix}$.

(B) $\begin{bmatrix} 3 & -4 \\ -4 & 2 \end{bmatrix}$.

(C) $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$.

(D) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

FIFTH SEMESTER U.G. DEGREE EXAMINATION, NOVEMBER 2021

(CBCSS—UG)

Mathematics

MTS 5B 09—INTRODUCTION TO GEOMETRY

(2019 Admissions)

Time : Two Hours

Maximum : 60 Marks

Section A

*Answer at least eight questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 24.*

- Find the equation of the tangent at the point with parameter t to the parabola with parametric equations $x = at^2$, $y = 2at$ where $t \in \mathbb{R}$.
- Let E be a parabola with parametric equations $x = t^2$, $y = t$, $t \in \mathbb{R}$. Find focus, vertex axis and directrix of E .
- Prove that the equation of the tangent at the point (x_1, y_1) to an ellipse is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$.
- Write the equation of the conic $x^2 - 4xy + 4y^2 - 6x - 8y + 5 = 0$ in matrix form.
- Show that $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ is orthogonal for each real number θ .
- Let the Euclidean transformations t_1 and t_2 of $\mathbb{R}^2$ be given by :

$$t_1(X) = \begin{bmatrix} \frac{3}{5} & \frac{-4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix} X + \begin{bmatrix} 1 \\ -2 \end{bmatrix} \text{ and}$$

$$t_2(X) = \begin{bmatrix} \frac{-4}{5} & \frac{3}{5} \\ \frac{3}{5} & \frac{4}{5} \end{bmatrix} x + \begin{bmatrix} -2 \\ 1 \end{bmatrix}. \text{ Find } t_2 \circ t_1.$$

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7. Find the inverse of the affine transformation $t(X) = \begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix} X + \begin{bmatrix} 4 \\ -2 \end{bmatrix}$.
8. State fundamental theorem of affine geometry.
9. Prove that an affine transformation maps straight lines to straight lines.
10. State Desargue's theorem.
11. Find the equation of the line that passes through the point $[1, 2, 3]$ and $[2, -1, 4]$.
12. Find the point of intersection of the lines in $\mathbb{RP}^2$ with equations $x + 6y - 5z = 0$ and $x - 2y + z = 0$.

(8 × 3 = 24 marks)

Section B

Answer at least five questions.

Each question carries 5 marks.

All questions can be attended.

Overall Ceiling 25.

13. Let PQ be an arbitrary chord of the ellipse with equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Let M be the midpoint of PQ. Prove that the following expression is independent of the choice of P and Q : Slope of OM × Slope of PQ.
14. State and prove reflection properties of the ellipse.
15. Prove that the set of all affine transformations $A(2)$ forms a group under the operation of composition of functions.
16. Determine the image of the line $y = 2x$ under the affine transformation $t(X) = \begin{pmatrix} 4 & 1 \\ 2 & 1 \end{pmatrix} X + \begin{pmatrix} 2 \\ -1 \end{pmatrix}, X \in \mathbb{R}^2$.
17. Determine the affine transformation which maps the points $(2, 3)$, $(1, 6)$ and $(3, -1)$ to the points $(1, -2)$, $(2, 1)$ and $(-3, 5)$ respectively.

18. Prove that affine transformations maps ellipses to ellipses, parabolas to parabolas and hyperbolas to hyperbolas.
19. Determine the point of $\mathbb{RP}^2$ at which the line through the points $[1, 2, -3]$ and $[2, -1, 0]$ meets the line through the points $[1, 0, -1]$ and $[1, 1, 1]$.

(5 × 5 = 25 marks)

Section C

Answer any one question.

The question carries 11 marks.

20. Prove that the conic E with equation $3x^2 - 10xy + 3y^2 + 14x - 2y + 3 = 0$ is a hyperbola. Determine its centre, and its major and minor axes.
21. State and prove Ceva's theorem.

(1 × 11 = 11 marks)

FIFTH SEMESTER U.G. DEGREE EXAMINATION, NOVEMBER 2021**(CBCSS—UG)****Mathematics****MTS 5B 08—LINEAR PROGRAMMING****(2019 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 15****Maximum : 15 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 15.
2. The candidate should check that the question paper supplied to him/her contains all the 15 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 5B 08—LINEAR PROGRAMMING

(Multiple Choice Questions for SDE Candidates)

1. "Any unbounded linear programming problem has an unbounded constraint set" :
 - (A) The above statement is always TRUE.
 - (B) The above statement is always FALSE.
 - (C) The above statement can be TRUE or FALSE.
 - (D) Insufficient data.
2. A linear programming problem having 6 main constraints and 3 non-negativity constraints. Find the upper bound for the number of extreme point candidates.
 - (A) 504.
 - (B) 120.
 - (C) 20.
 - (D) 84.
3. Which of the following is/are non-convex set(s) ?
 - (i) x -axis, (ii) $\mathbb{R}^2 - \{(0,0)\}$, (iii) Unit Circle.
 - (A) All the above.
 - (B) (i), (iii) but not (ii).
 - (C) (ii) and (iii) but not (i).
 - (D) Only (i).
4. The variables to the south of the minimum tableau are called :
 - (A) Basic variables.
 - (B) Slack variables.
 - (C) Non-basic variables.
 - (D) Independent variables.
5. A pivot entry is :
 - (A) Always negative.
 - (B) Always positive.
 - (C) Always zero.
 - (D) Always non-zero.
6. In non-canonical maximum tableaux :
 - (A) Rows corresponding to pivoted unconstrained variables are filed and deleted.
 - (B) Rows corresponding to pivoted unconstrained variables are filed and never deleted.
 - (C) Columns corresponding to pivoted unconstrained variables are filed and deleted.
 - (D) Columns corresponding to pivoted unconstrained variables are filed and never deleted.

7. Slack “variables” of 0 (corresponding to equations of constraint) in non-canonical maximum tableaus always :
- (A) Get pivoted up-from east to north.
 - (B) Get pivoted up-from north to east.
 - (C) Get pivoted up-from west to south.
 - (D) Get pivoted up-from south to west.
8. Unconstrained independent variables in non-canonical maximum tableaus always :
- (A) Get pivoted down-from north to east.
 - (B) Get pivoted down-from south to west.
 - (C) Get pivoted down-from west to south.
 - (D) Get pivoted down-from east to north.
9. “In a minimum basic feasible tableau, the basic solution is a feasible solution” :
- (A) FALSE.
 - (B) TRUE.
 - (C) May be TRUE.
 - (D) May be FALSE.
10. Given dual canonical linear programming problem if Maximization problem has optimal solution then Minimization problem has :
- (A) Unbounded solution.
 - (B) Infeasible solution.
 - (C) Unbounded solution or infeasible solution.
 - (D) Optimal solution.
11. In L.P.P. unbounded solution means :
- (A) Infeasible solution.
 - (B) Degenerate Solution.
 - (C) Infinite solution.
 - (D) Unique Solution.
12. Assignment problem is a special type of transportation problem :
- (A) Always true.
 - (B) May be true.
 - (C) Always false.
 - (D) May be false.

13. To find initial feasible solution of a transportation problem, the method which starts allocation from the minimum cost is called _____.
- (A) Minimum entry method.
 - (B) Northwest corner method.
 - (C) Northeast corner method.
 - (D) VAM method.
14. If the number of rows and columns in an assignment problem are not equal then it is said to be :
- (A) Bounded.
 - (B) Infeasible.
 - (C) Unbounded.
 - (D) Unbalanced.
15. If there are n workers and n jobs and each worker assign to only one job then there would be :
- (A) $n!$ solutions.
 - (B) $(n - 1)!$ solutions.
 - (C) $(n + 1)!$ solutions.
 - (D) n solutions.